

The maximum entropy principle meets optimality theory

Computational modeling of language
Spring 2026

Bureaucracy

- Midterm: scores have been **adjusted**.
- See Iñigo for your exam (**please keep private**), and for discussion.
- If your score on the final exam is higher than on the midterm, I will **drop** your midterm from consideration.

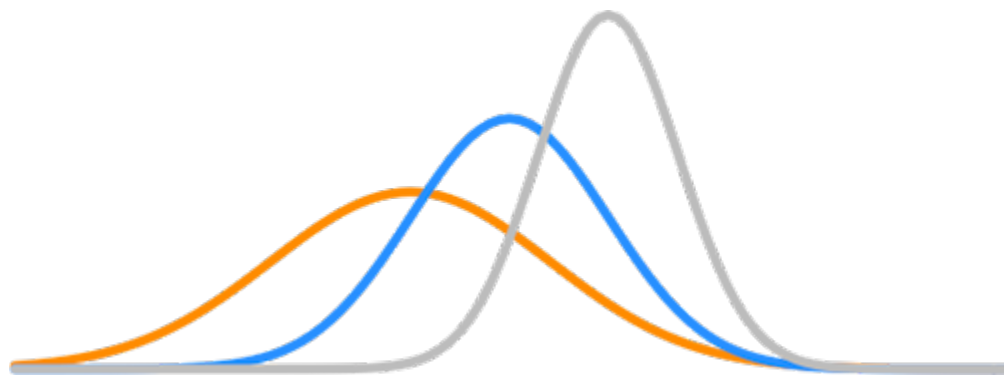
Overview

1. Maximum entropy
2. Optimality theory

Overview

1. Maximum entropy

2. Optimality theory



More statistical learning

Maximum entropy

Maximum entropy

Q: What is the principle of maximum entropy?

Information theory



Maximum entropy

Information theory



Maximum entropy

$\approx$

Logistic regression

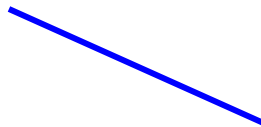
Information theory



Maximum entropy

$\approx$

Logistic regression



Naive Bayes

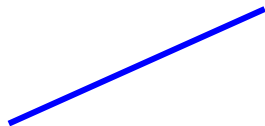
Information theory



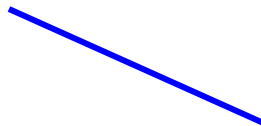
Maximum entropy

$\approx$

Logistic regression



Neural networks



Naive Bayes

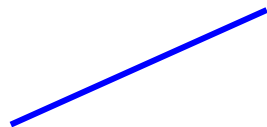
Information theory



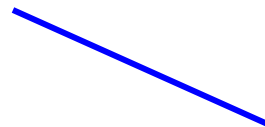
Maximum entropy

$\approx$

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Naive Bayes

Q: Where do other things we've learned fit in this map?

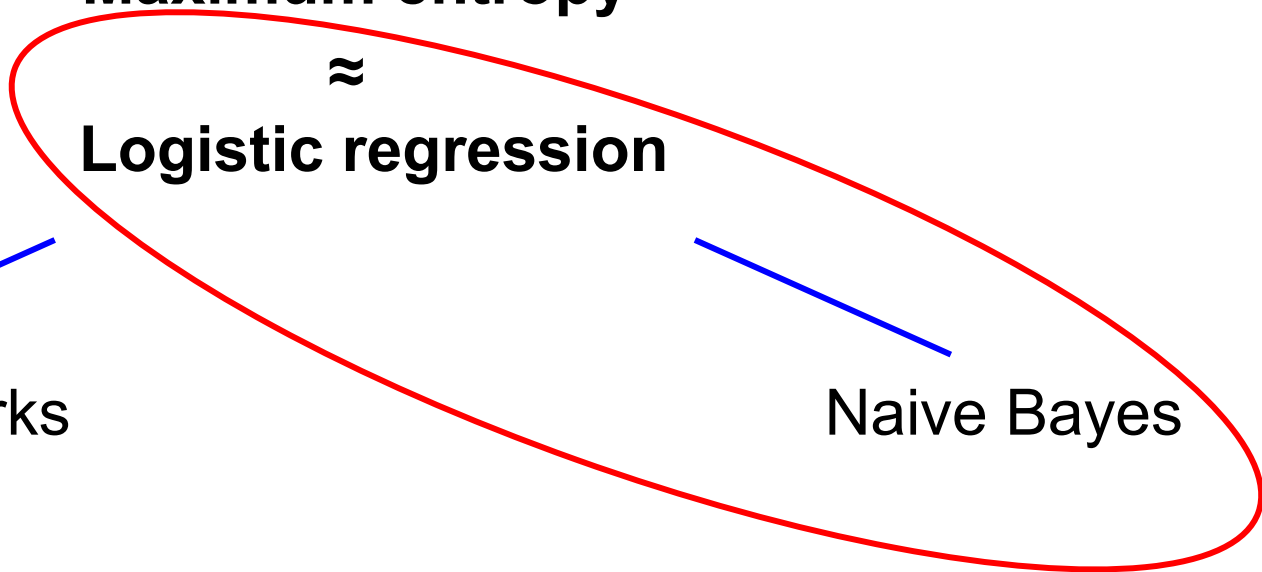
Information theory



Maximum entropy

$\approx$

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Naive Bayes

THE SIMPLEST SPAM-FILTER

(used until 2010)

hey	...	1829
I'm	...	1710
no	...	1191
where	...	1012
you	...	985
speak	...	873
learn	...	747
one	...	739

good letters

672 times

«KITTY»

13 times

viagra	...	1552
casino	...	1492
100%	...	1320
credit	...	1184
sale	...	985
press	...	873
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spam letters

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

BAYES' THEOREM



NOT SPAM

NAIVE BAYES

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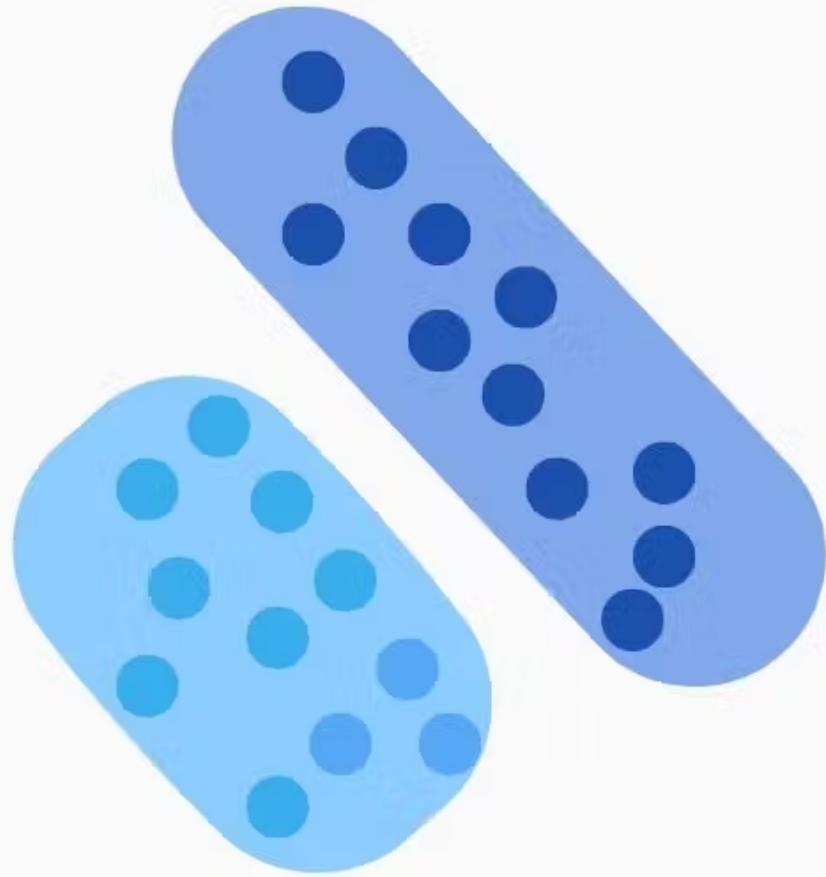
spam letters

$$P(Y = 1|X) = \frac{1}{1 + \exp(w_0 + \sum_{i=1}^n w_i X_i)}$$



NOT SPAM

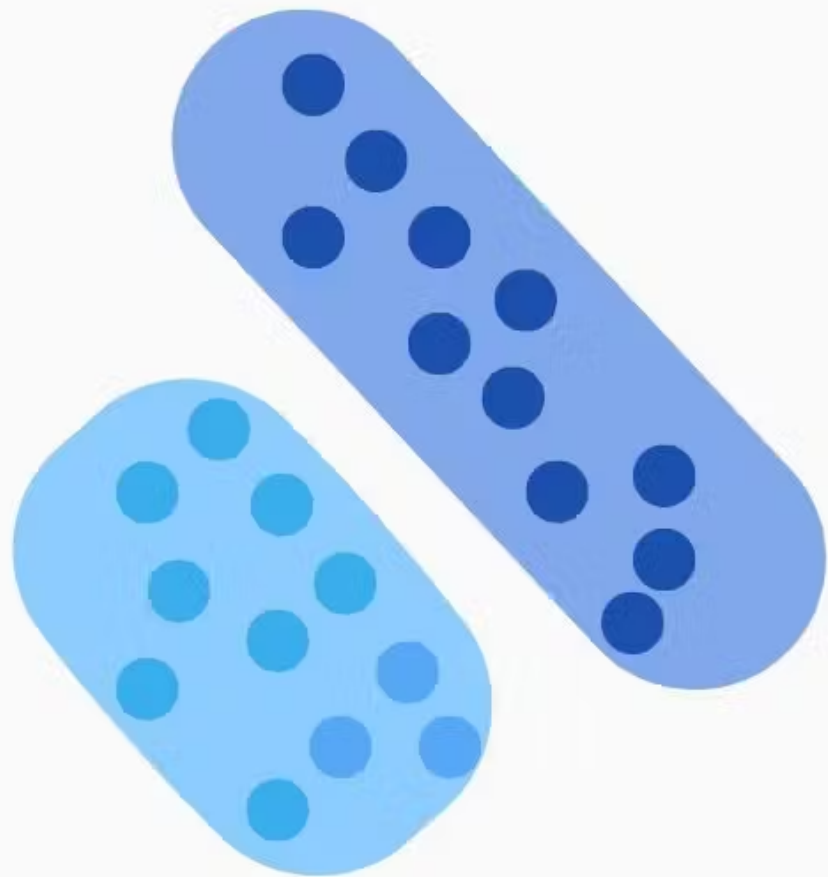
Logistic regression



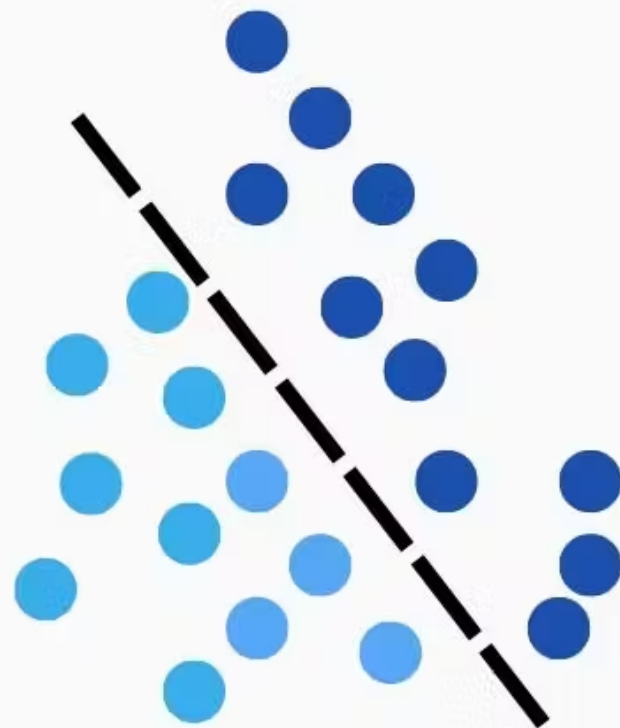
Generative

Generative classifier
(as in generative AI)

Classifies by modeling
**how the data were
generated**: there is a
term for $P(\text{data}|\text{class})$.
E.g. naive Bayes.



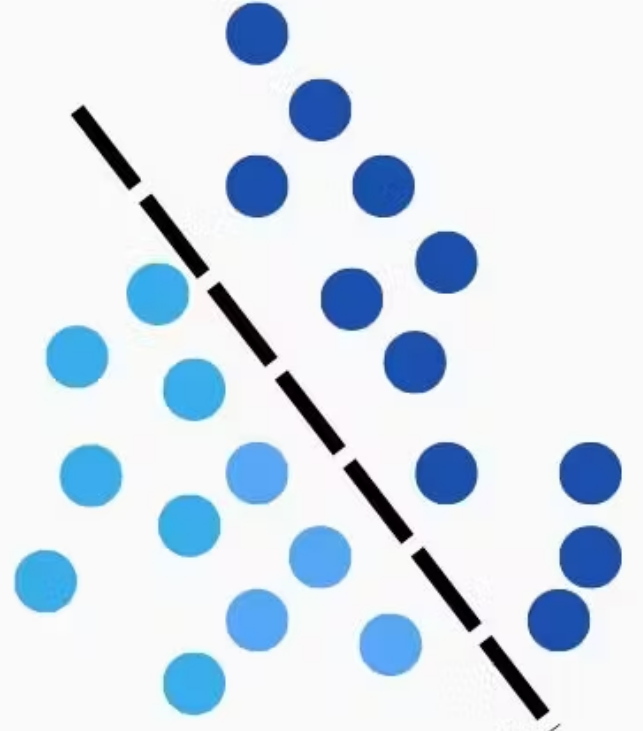
Generative



Discriminative

Discriminative classifier

Classifies by modeling **boundaries**, without caring how the data were generated: **there is no term for $P(\text{data}|\text{class})$** .
E.g. logistic regression.



Discriminative

Naive Bayes

If we want to predict a class y (e.g. spam vs. non-spam) from a set of features $x_1, \dots, x_n$ (e.g. words in a document), then Bayes' rule says:

$$\underbrace{P(y \mid x_1, \dots, x_n)}_{\text{posterior}} \propto \underbrace{P(y)}_{\text{prior}} \underbrace{P(x_1, \dots, x_n \mid y)}_{\text{likelihood}}$$

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$$\underbrace{P(y \mid x_1, \dots, x_n)}_{\text{posterior}} \propto \underbrace{P(y) \underbrace{P(x_1, \dots, x_n \mid y)}_{\text{likelihood}}}_{\text{Generative classifier}}$$

Logistic regression

$$P(Y = 1|X) = \frac{1}{1 + \exp(w_0 + \sum_{i=1}^n w_i X_i)}$$


Discriminative classifier: no term for $P(X|Y)$

Logistic regression as a pipeline

features

$$x_0$$

⋮

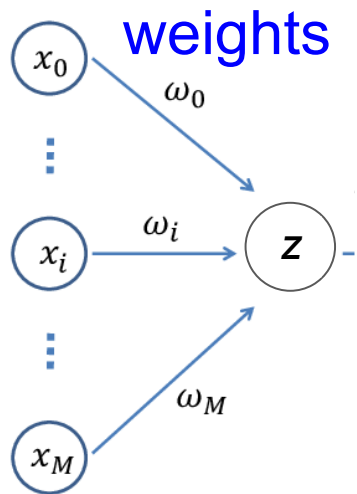
$$x_i$$

⋮

$$x_M$$

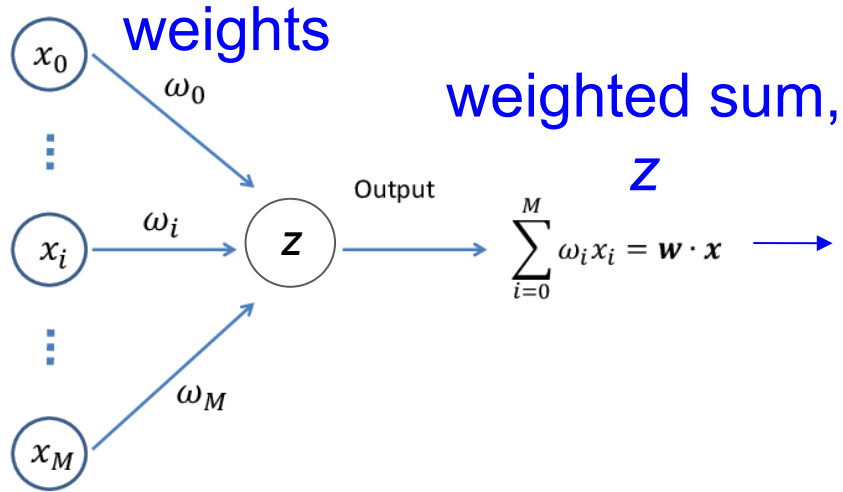
Logistic regression as a pipeline

features



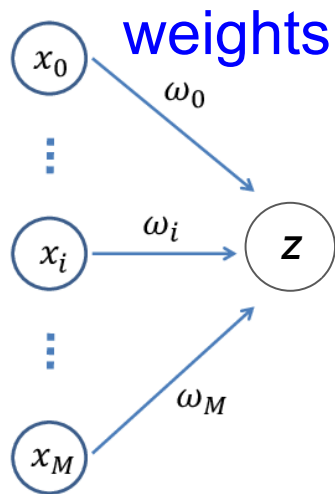
Logistic regression as a pipeline

features



Logistic regression as a pipeline

features



weighted sum,

z

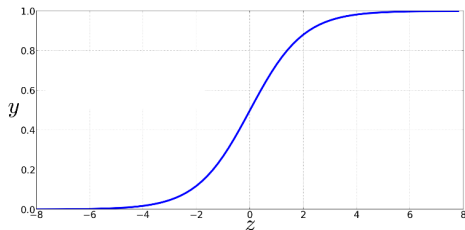
Output

$$\sum_{i=0}^M \omega_i x_i = \mathbf{w} \cdot \mathbf{x}$$

→

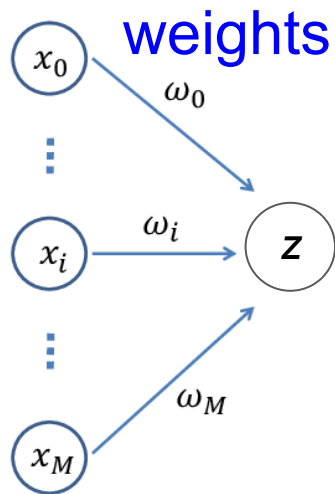
sigmoid

$$y = \sigma(z) = \frac{1}{1 + e^{-z}}$$



Logistic regression as a pipeline

features



weighted sum,

z

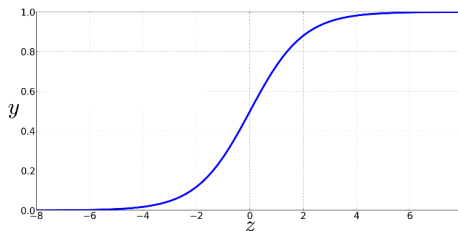
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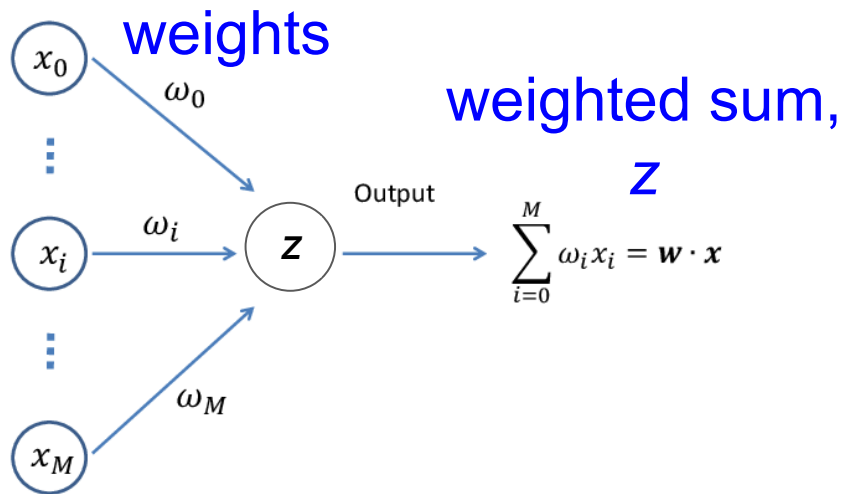
classification
probability



→ $P(y = 1 \mid x)$

Wait! Where's the bias term b ?

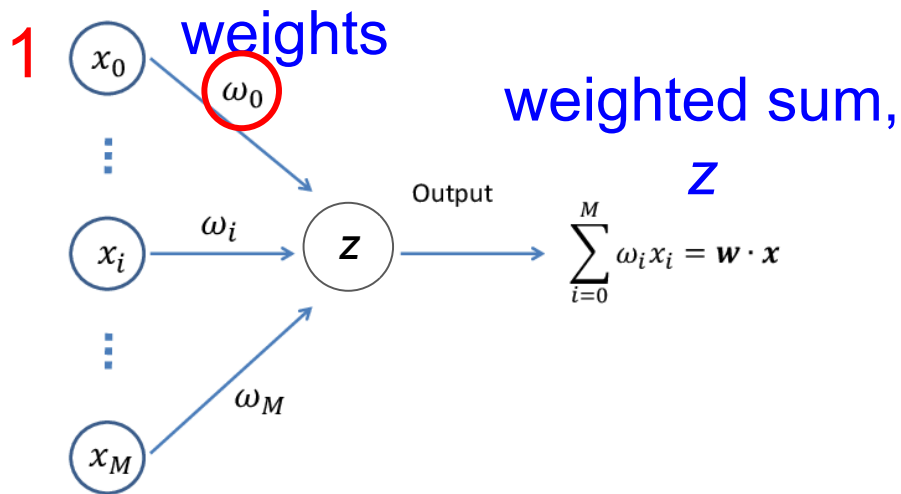
features



$$z = \left(\sum_{i=1}^n w_i x_i \right) + b$$

Wait! Where's the bias term b ?

features



$$z = \left(\sum_{i=1}^n w_i x_i \right) + b$$

Folded into weight vector.
 b = weight w_0 for dummy
feature x_0 which always
has value 1.

Multinomial logistic regression (more than two classes)

$$p(y = c|x) \propto e^{w_c \cdot x + b_c}$$

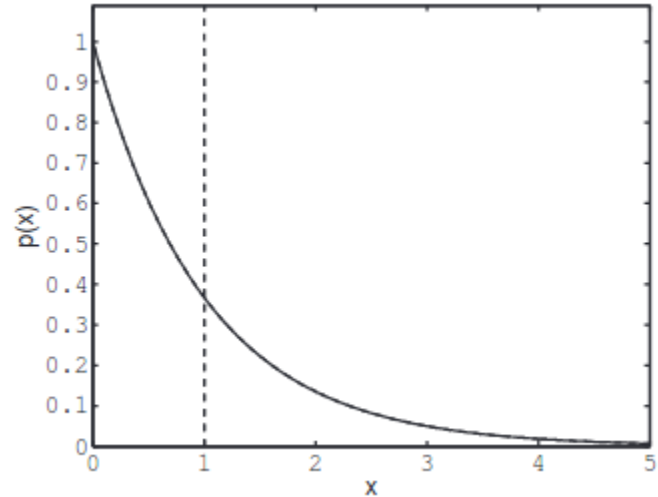
Multinomial logistic regression (more than two classes)

$$p(y = c|x) \propto e^{w_c \cdot x + b_c}$$

Maxent classifier. This is a **generalization** of the 2-class formula, to multiple classes. The 2-class formula is a **special case** of this.

Example 2: If a variable x has **no values below 0**, and has a fixed **mean** μ , but is otherwise unconstrained, then the maximum entropy distribution is exponential:

$$p(x) = \frac{1}{\mu} e^{-x/\mu}$$



General maxent classifier: $P(y = 1|x) \propto \exp(w_1 \cdot x)$

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$$\begin{aligned} P(y = 1|x) &\propto \exp(w_1 \cdot x) \\ &= \frac{\exp(w_1 \cdot x)}{\exp(w_1 \cdot x) + \exp(w_0 \cdot x)} \end{aligned}$$

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Standard 2-class logistic regression:

where $w = w_1 - w_0$

Learning

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- Behavior of classifier depends on weights.

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- Learning: adjusting weights so that classifier provides the “right answers”.
- The “right answer” $y=c$ for a given input x is specified in a training set.
- So: learning is adjusting weights to maximize the probability $p(y=c|x)$, for each input/output pair $\{x, y=c\}$ in the training set.

Learning requires

1. A **loss function** $L(y, y^*)$ capturing how far our actual output y is from our desired output y^* .
E.g. cross-entropy loss. In general, **reducing loss** should **increase probability** of the correct answer.

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E.g. cross-entropy loss. In general, **reducing loss** should **increase probability** of the correct answer.
2. An **optimization method** for adjusting weights to reduce loss, e.g. gradient descent.

Cross-entropy loss

Given a true probability distribution P (e.g. desired output), and another distribution Q that is meant to approximate it (e.g. actual output), the **cross-entropy** between P and Q is:

$$H(P, Q) = - \sum_{x \in \mathcal{X}} p(x) \log q(x)$$

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Cross-entropy can be seen as a measure of **divergence** of Q from P . $H(P, Q)$ is **minimized** [at value $H(P)$] when $Q = P$.

Relation to KL divergence

Kullback-Leibler (KL) divergence of an estimate distribution Q from a true distribution P is:

$$D_{KL}(P \parallel Q) = \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{q(x)}$$

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So cross-entropy is **the same as** KL divergence except for an additive constant, $H(P)$!

Cross-entropy **links** two quantities we already know: entropy and KL divergence.

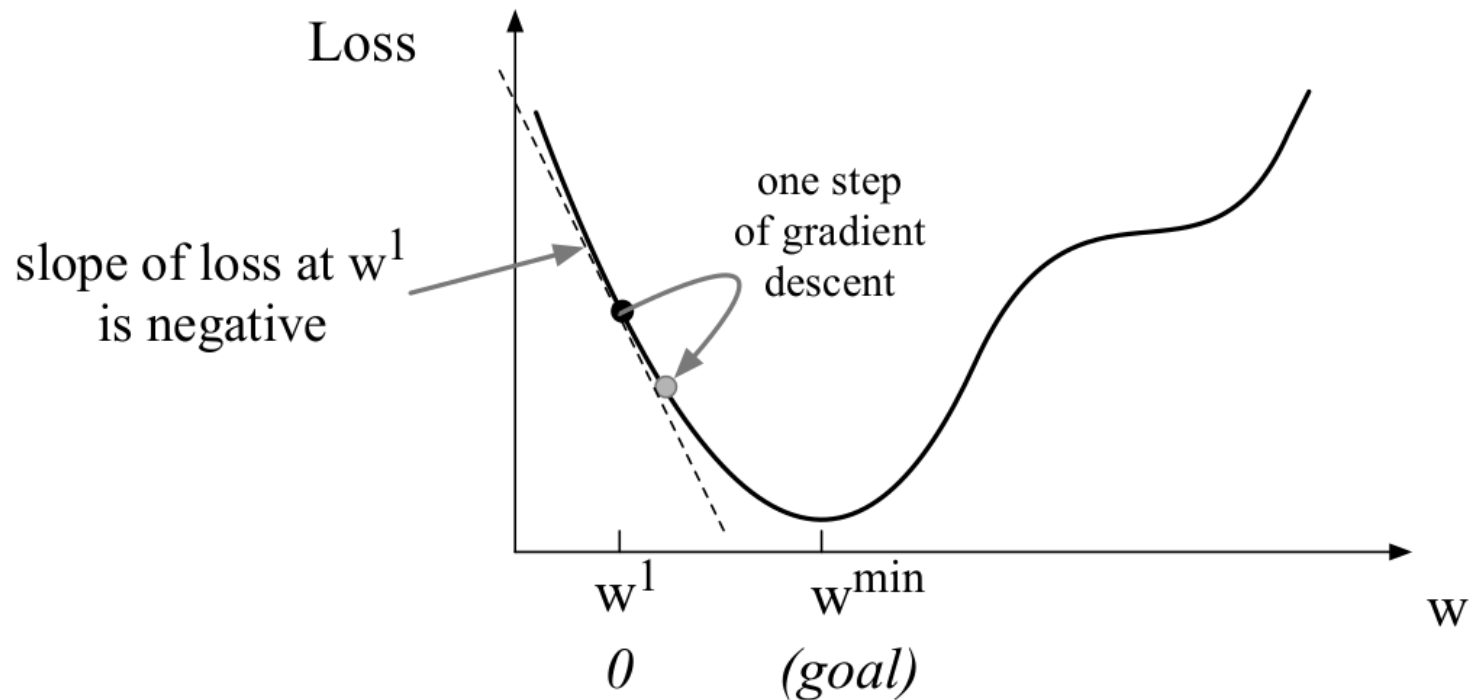


Figure 5.3 The first step in iteratively finding the minimum of this loss function, by moving w in the reverse direction from the slope of the function. Since the slope is negative, we need to move w in a positive direction, to the right. Here superscripts are used for learning steps, so w^1 means the initial value of w (which is 0), w^2 at the second step, and so on.

An interactive tutorial on numerical optimization

It's good to understand gradient descent and related optimization methods, but please know that:

`scipy.optimize.minimize` is your friend!

Discussion

What are the **pros** and **cons**, pedagogically and from a student's perspective, of using pre-fab routines like `scipy.optimize.minimize`, `sklearn.naive_bayes.BernoulliNB`, etc. — as opposed to coding them from scratch?

I've already given you my view of this but am curious to hear yours — especially **cons**.

Overview

1. Maximum entropy

2. Optimality theory

Optimality theory (OT)

In linguistics, **Optimality Theory** ... [holds] that the observed forms of language arise from the **optimal satisfaction of conflicting constraints**. ... OT models grammars as systems that provide mappings from inputs to outputs; typically, the inputs are conceived of as underlying representations, and the outputs as their surface realizations. It is an approach within the larger framework of generative grammar.

- Shamelessly copied from [Wikipedia](#)

Constraints, strictly ranked

Input	iti	*[rtr / hi]	PARSE (rtr)	*GESTURE (contour)	PARSE (atr)	*[atr / lo]
Outputs	 iti					
		*!		*	*	
	iti	*!		*	*	
	iti	*!*			**	

An optimality theory (OT) **tableau**

Tableau

Input	CONSTRAINT 1	CONSTRAINT 2	CONSTRAINT 3
a.  Candidate A	*	*	***
b. Candidate B	*	**!	

* : a violation of this constraint by this candidate

! : fatal violation - candidate no longer viable

pointing finger : winning candidate output

Type	Name	Description
Markedness	*SS	Two successive sibilants are prohibited. One violation for every pair of adjacent sibilants in the output.
	AGREE(Voice)	Output segments agree in specification of [±voice] . One violation for every pair of adjacent obstruents in the output which disagree in voicing.
Faithfulness	MAX	Maximizes all input segments in the output. One violation for each segment in the input that does not appear in the output. This constraint prevents deletion.
	DEP	Output segments are dependent on having an input correspondent. One violation for each segment in the output that does not appear in the input. This constraint prevents insertion.
	IDENT(Voice)	Maintains the identity of the [±voice] specification. One violation for each segment that differs in voicing between the input and output.

/dɒg/ + /z/ → [dɒgz]

/dɒg/ + /z/	*SS	AGREE	MAX	DEP	IDENT
a.  dɒgz					
b. dɒgs		*!			*
c. dɒgɪz				*!	
d. dɒgɪs				*!	*
e. dɒg			*!		

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/dɒg/ + /z/ → [dɒgz]

/dɒg/ + /z/	*SS	AGREE	MAX	DEP	IDENT
a. dɒgz					
b. dɒgs		*!			*
c. dɒgɪz				*!	
d. dɒgɪs				*!	*
e. dɒg			*!		

/kæt/ + /z/ → [kæts]

/kæt/ + /z/	*SS	AGREE	MAX	DEP	IDENT
a. kætɪz		*!			
b. kæts					*
c. kætɪz				*!	
d. kætɪs				*!	*
e. kæt			*!		

Type	Name	Description
Markedness	*SS	Two successive sibilants are prohibited. One violation for every pair of adjacent sibilants in the output.
	AGREE(Voice)	Output segments agree in specification of [±voice] . One violation for every pair of adjacent obstruents in the output which disagree in voicing.
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/dɒg/ + /z/ → [dɒgz]

/dɒg/ + /z/	*SS	AGREE	MAX	DEP	IDENT
a.  dɒgz					
b. dɒgs		*!			*
c. dɒgɪz				*!	
d. dɒgɪs				*!	*
e. dɒg			*!		

/kæt/ + /z/ → [kæts]

/kæt/ + /z/	*SS	AGREE	MAX	DEP	IDENT
a. kætɪz		*!			
b.  kæts					*
c. kætɪz				*!	
d. kætɪs				*!	*
e. kæt			*!		

/dɪf/ + /z/ → [dɪfɪz]

/dɪf/ + /z/	*SS	AGREE	MAX	DEP	IDENT
a. dɪfɪz	*!	*			
b. dɪfs	*!				*
c.  dɪfɪz				*	
d. dɪfɪs				*	*!
e. dɪf			*!		

Challenges

- How to **learn from data**? Either for a child, or for a scientist. Challenging to do by hand, especially from potentially noisy data.

Challenges

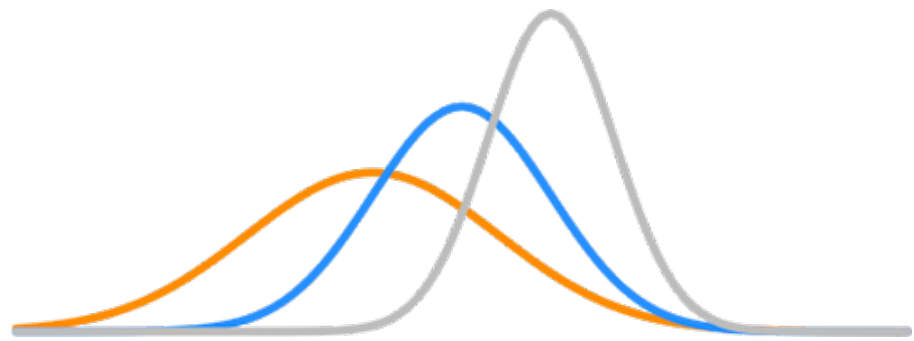
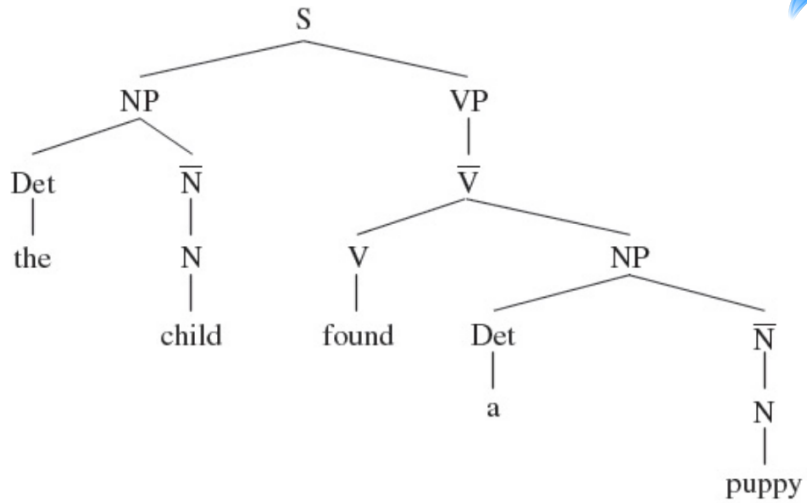
- How to **learn from data**? Either for a child, or for a scientist. Challenging to do by hand, especially from potentially noisy data.
- “**Grammars leak**” (Sapir). There are exceptions, e.g. **free variation** in which a single input form has more than one grammatical output form.

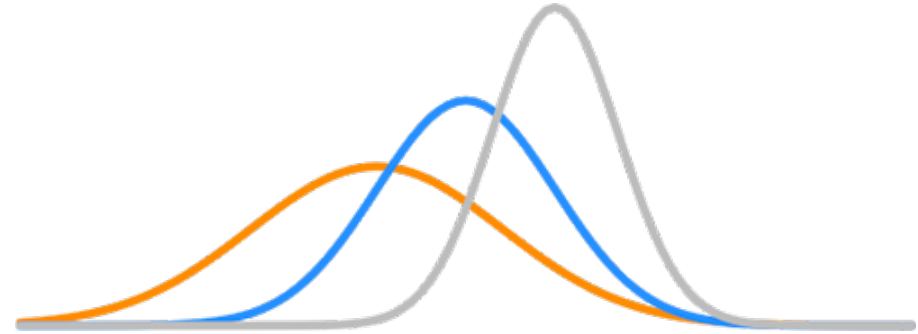
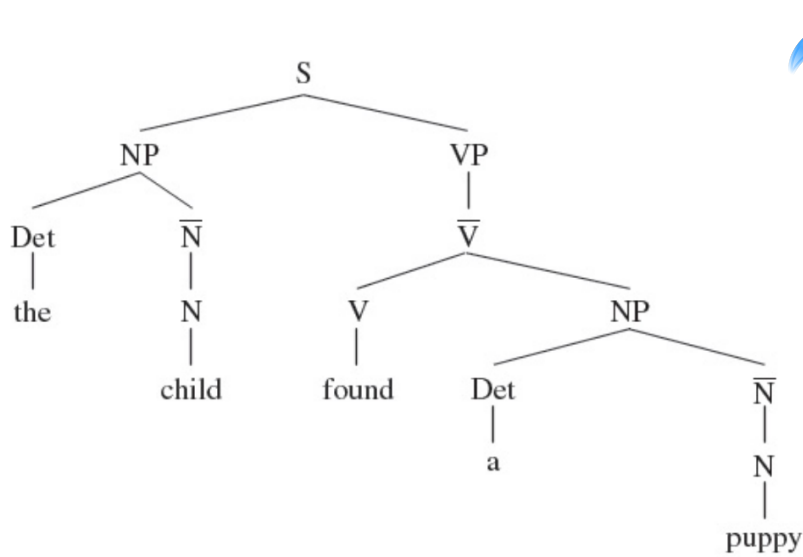
Learning OT Constraint Rankings Using a Maximum Entropy Model

Sharon Goldwater and Mark Johnson

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Abstract. A weakness of standard Optimality Theory is its inability to account for grammars with free variation. We describe here the Maximum Entropy model, a general statistical model, and show how it can be applied in a constraint-based linguistic framework to model and learn grammars with free variation, as well as categorical grammars. We report the results of using the MaxEnt model for learning two different grammars: one with variation, and one without. Our results are as good as those of a previous probabilistic version of OT, the Gradual Learning Algorithm (Boersma, 1997), and we argue that our model is more general and mathematically well-motivated.





- Maxent OT is a salient point of contact between **generative linguistics** and **statistical learning**.

Constraints

Input x

Outputs y

iti	*[rtr / hi]	PARSE (rtr)	*GESTURE (contour)	PARSE (atr)	*[atr / lo]
 iti					
iti	*!		*	*	
iti	*!		*	*	
iti	*!*			**	

Constraints

Input x

Outputs y

	iti	*[rtr / hi]	PARSE (rtr)	*GESTURE (contour)	PARSE (atr)	*[atr / lo]
{	 iti					
	iti	*!		*	*	
	iti	*!		*	*	
	iti	*!*			**	

Constraint violations $f_i(y,x)$

OT in terms of maximum entropy (MaxEnt OT)

$$\Pr(y|x) \propto \exp\left(\sum_{i=1}^m w_i f_i(y, x)\right)$$

OT in terms of maximum entropy (MaxEnt OT)

Output **Input** **Constraint weights**

$$\Pr(\overline{y}|\overline{x}) \propto \exp\left(\sum_{i=1}^m \overline{w_i} \underbrace{f_i(\overline{y}, \overline{x})}_{\text{Constraint violations}}\right)$$

Constraint violations

Constraints

Input x

Outputs y

Constraints					
iti	*[rtr / hi]	PARSE (rtr)	*GESTURE (contour)	PARSE (atr)	*[atr / lo]
 iti					
iti	*!		*	*	
iti	*!		*	*	
iti	*!*			**	

Constraint violations $f_i(y,x)$

Adjust weights to maximize probability of **winner**.

Maxent OT loss function

$$\log \text{PL}_{\bar{w}}(\bar{y}|\bar{x}) - \sum_{i=1}^m \frac{(w_i - \mu_i)^2}{2\sigma_i^2}$$

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$$\log \text{PL}_{\bar{w}}(\bar{y}|\bar{x}) - \sum_{i=1}^m \frac{(w_i - \mu_i)^2}{2\sigma_i^2}$$

Maximize the probability of the intended winning output,
for each input.

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Maximize the probability of the intended winning output, for each input.

$$\text{PL}_{\bar{w}}(\bar{y}|\bar{x}) = \prod_{j=1}^n \text{Pr}_{\bar{w}}(Y = \underline{y_j} | x(Y) = x_j)$$

winning output

Maxent OT loss function

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[Link to cross-entropy loss!](#)
winning output

Maxent OT and cross-entropy loss

The **cross-entropy** between desired output P and actual output Q is:

$$H(P, Q) = - \sum_{x \in \mathcal{X}} p(x) \log q(x)$$

Maxent OT and cross-entropy loss

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$$- \log q(x^*)$$

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Minimizing this is the same as **maximizing** the probability of the winning output, in Goldwater & Johnson's formulation.

Maxent OT loss function

$$\log \text{PL}_{\bar{w}}(\bar{y}|\bar{x}) - \sum_{i=1}^m \frac{(w_i - \mu_i)^2}{2\sigma_i^2}$$

Regularizing term: intended to **prevent overfitting**. We assume that $\mu_i = 0$, $\forall i$, and σ_i has the same value σ , $\forall i$.

Categorical grammar: Wolof

*RTRHi>>PARSE[RTR]>>GESTURE[CONTOUR]>>PARSE[ATR]>>*ATRLo

Constraint	Weight
*RTRHi	33.89
PARSE[RTR]	17.00
GESTURE[CONTOUR]	10.00
PARSE[ATR]	3.53
*ATRLo	0.41

Categorical grammar: Wolof

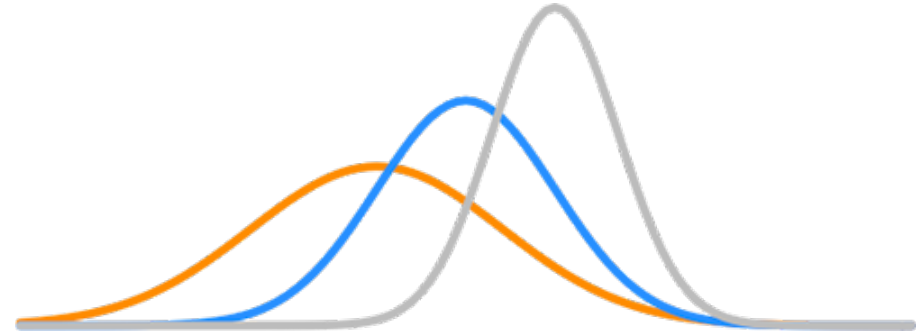
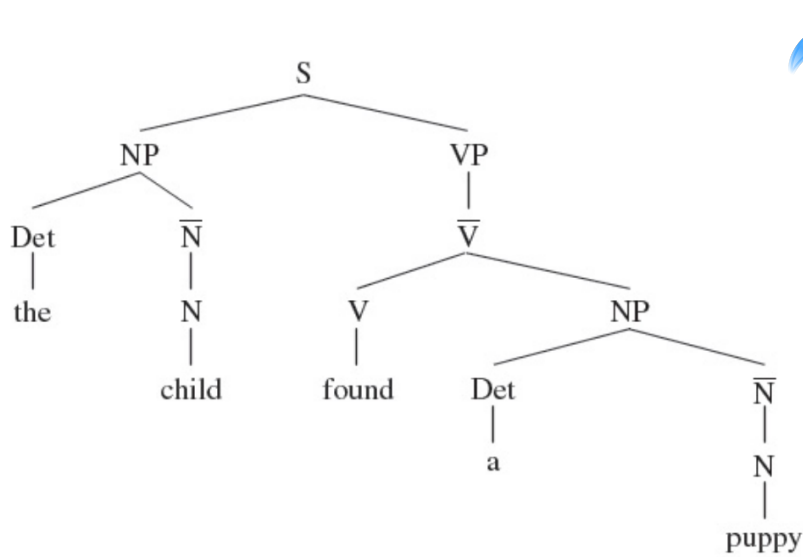
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Constraint	Weight
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PARSE[RTR]	17.00
GESTURE[CONTOUR]	10.00
PARSE[ATR]	3.53
*ATRLo	0.41

Our homework will replicate this.

Stochastic grammar: Finnish

Class	Tokens	% Majority	GLA	MaxEnt
1	1097	100	99.5	99.6
2	1000	100	100.0	100.0
3	923	100	100.0	100.0
4	873	70.7	69.5	69.4
5	821	98.4	100	99.8
6	457	99.6	99.4	98.0
7	436	82.1	81.6	80.5
8	91	50.5	58.0	55.3



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Discussion

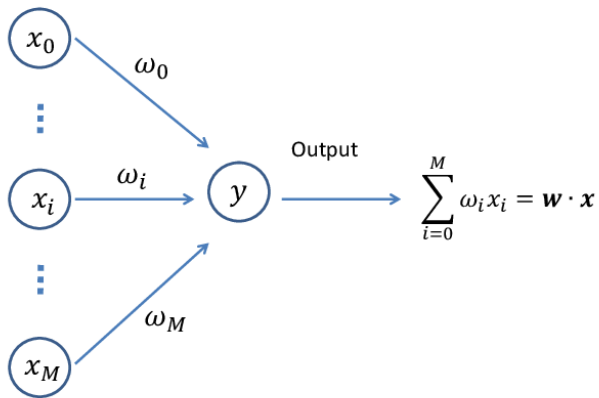
We have seen theoretical principles that were originally advanced within linguistics (OT) be elaborated by more general principles that originated outside linguistics (maximum entropy classification / multinomial logistic regression).

Is this a good thing? A bad thing? Neutral?
Why?

Questions?

Comments?

$$\Pr(y|x) = \frac{1}{Z(x)} \exp\left(\sum_{i=1}^m w_i f_i(y, x)\right)$$



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